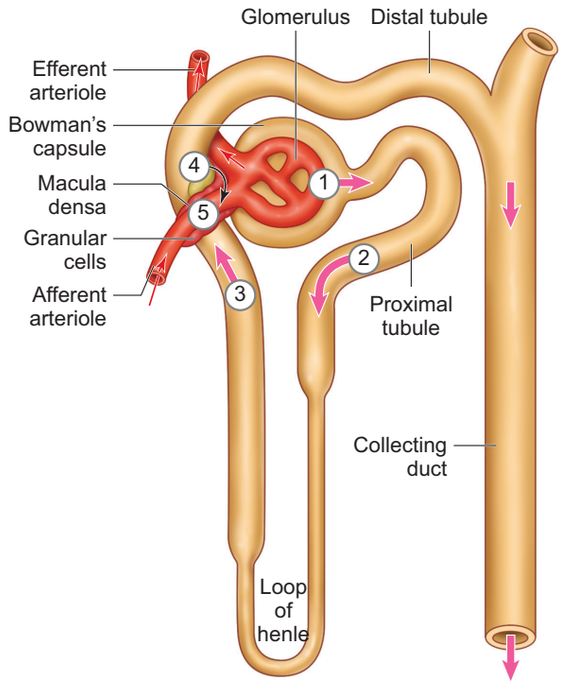


Figure 8.4 Structure of a Nephrons

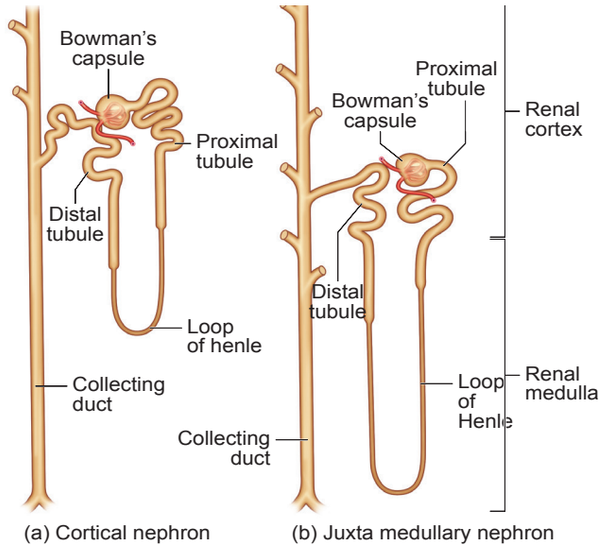


Figure 8.5 (a) Cortical nephrons are located predominantly in the outer cortex. (b) Juxtamedullary nephrons are mainly located in the inner medulla.

The renal tubule continues further to form the proximal convoluted tubule [PCT] followed by a U-shaped loop of Henle (Henle's loop) that has a thin descending and a thick ascending limb. The ascending limb continues as a highly coiled tubular region called the distal convoluted tubule [DCT]. The DCT of many nephrons open into a straight tube called collecting duct. The collecting duct runs through the medullary pyramids in the region of the pelvis. Several collecting ducts fuse to form papillary duct that delivers urine into the calyces, which opens into the renal pelvis.

In the renal tubules, PCT and DCT of the nephron are situated in the cortical region of the kidney whereas the loop of Henle is in the medullary region. In majority of nephrons, the loop of Henle is too short and extends only very little into the medulla and are called **cortical nephrons.** Some nephrons have very long loop of Henle that run deep into the medulla and are called **juxta medullary nephrons (JMN)** (Figure 8.5 a and b)

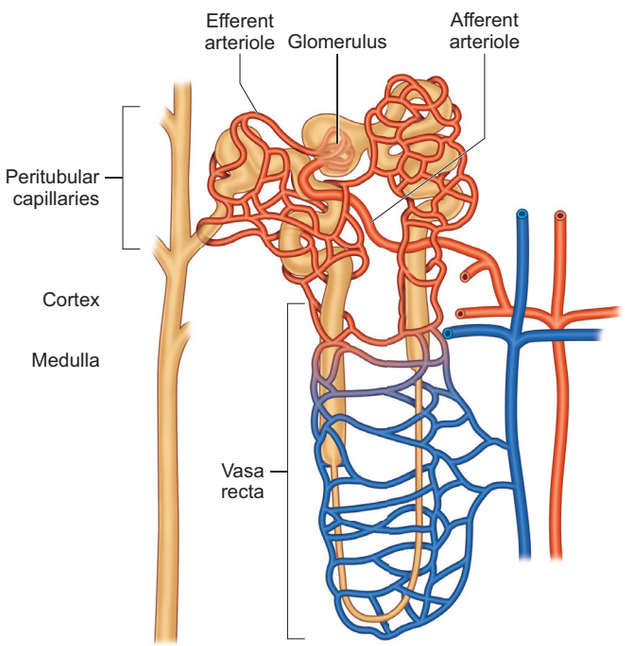


Figure 8.6 Blood vessels of the nephron.

The capillary bed of the nephrons- First capillary bed of the nephron is the glomerulus and the other is the peritubular capillaries. The glomerular capillary bed is different from other capillary beds in that it is supplied by the afferent and drained by the efferent arteriole. The efferent

TN_GOVT_XI_Bio Zoology_Ch08.indd 136 09-01-2020 5.29.56 PM